

SecureVault — Core Error Handling and Validation

-*Cipher Syndicate*

SecureVault implements multiple layers of validation, exception handling, and security checks to ensure stability, prevent unauthorized access, and safely handle unexpected user behaviour without crashing the application. Below are a few of the many examples:

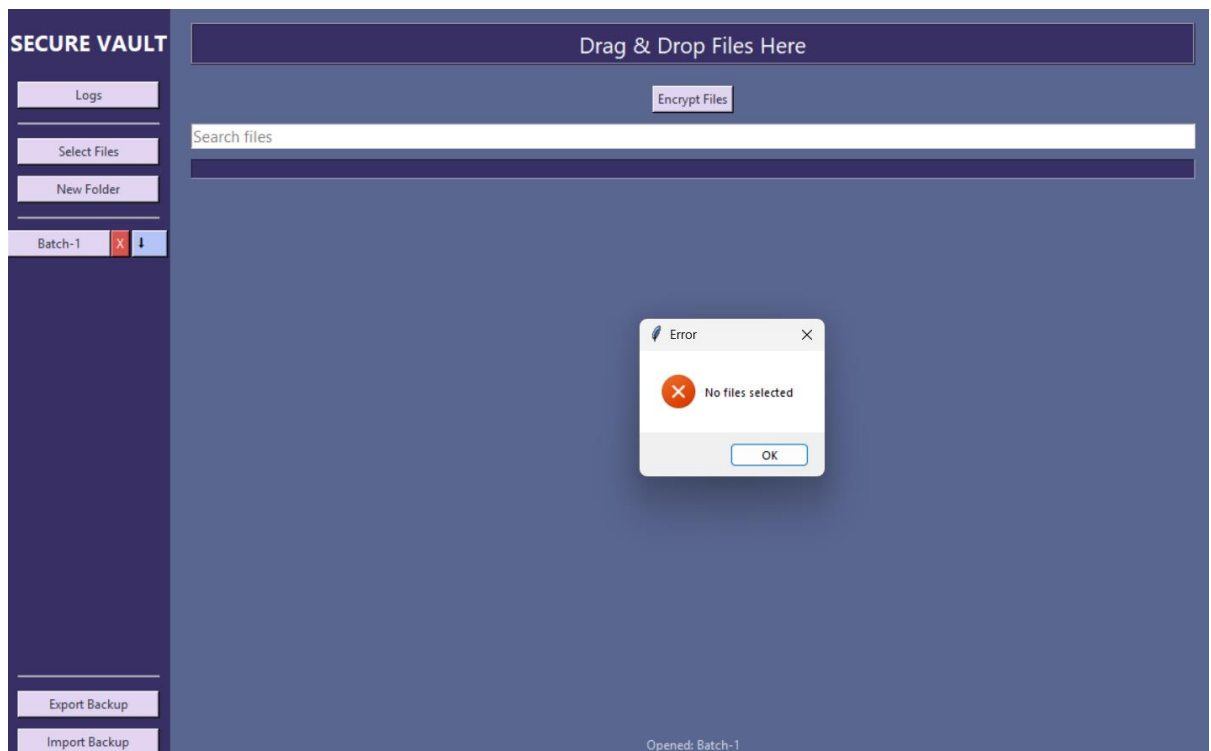
1. Empty File Selection Validation

- If a user attempts to encrypt files without selecting any files, the application safely stops the process, and informs the user instead of running valid operations.

- **Validation Logic:**

```
if not selected_files:
    messagebox.showerror(
        "Error",
        "No files selected"
    )
    return
```

- **Output Preview:**



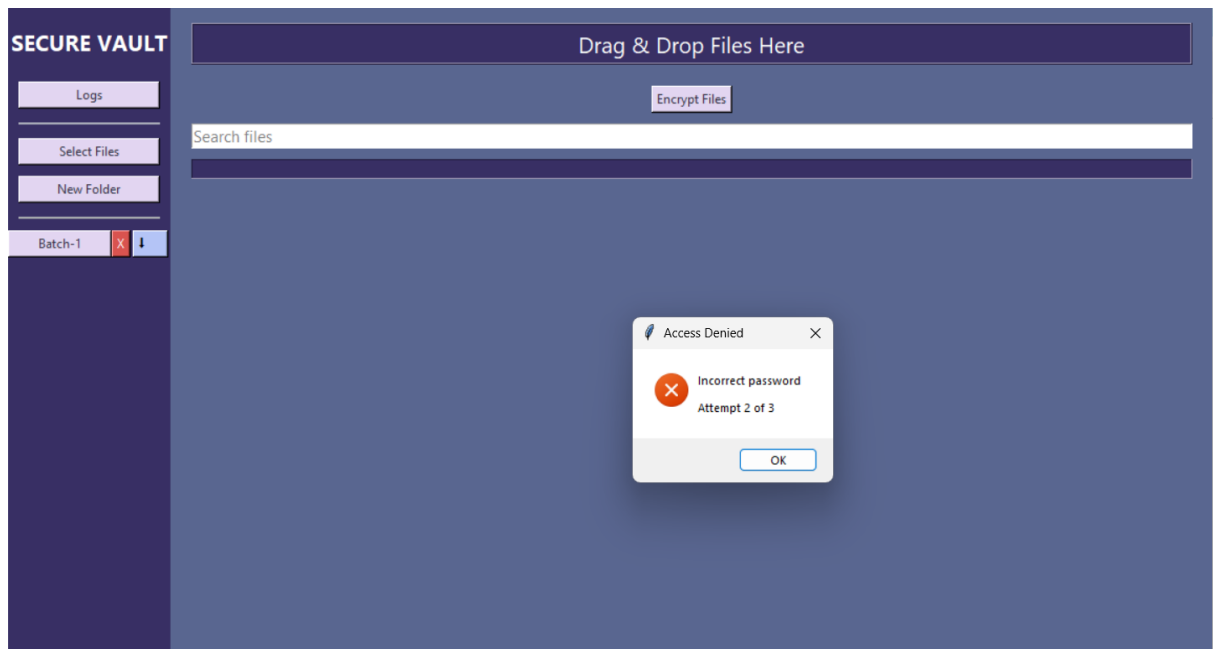
2. Password Authentication Validation

- If a user enters an incorrect vault password, unauthorized access attempts are blocked and logged.

- **Validation Logic:**

```
if not verify_folder_password(folder, pwd):  
  
    attempts = record_failed_attempt(  
        folder,  
        log_action,  
        LOGS_DIR  
    )  
  
    messagebox.showerror(  
        "Access Denied",  
        f"Incorrect password\n\nAttempt {attempts} of 3"  
    )  
  
    return
```

- **Output Preview:**



3. Intrusion Detection & Temporary Vault Lockout

- Repeated incorrect password attempts may indicate a brute-force attack; hence the vault temporarily locks itself after repeated failed attempts, preventing brute-force access.
- **Validation Logic:**

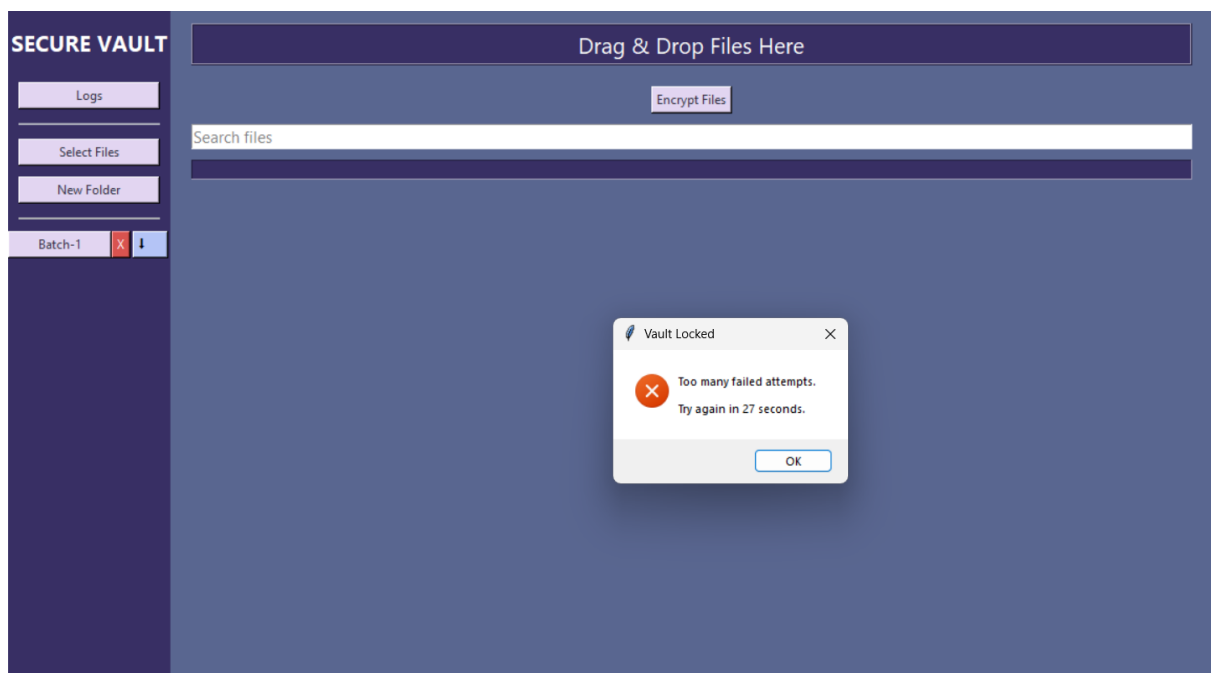
```
locked, remaining = is_folder_locked(folder)

if locked:

    messagebox.showerror(
        "Vault Locked",
        f"Too many failed attempts.\n\nTry again in {remaining}
seconds."
    )

    return
```

- **Output Preview:**



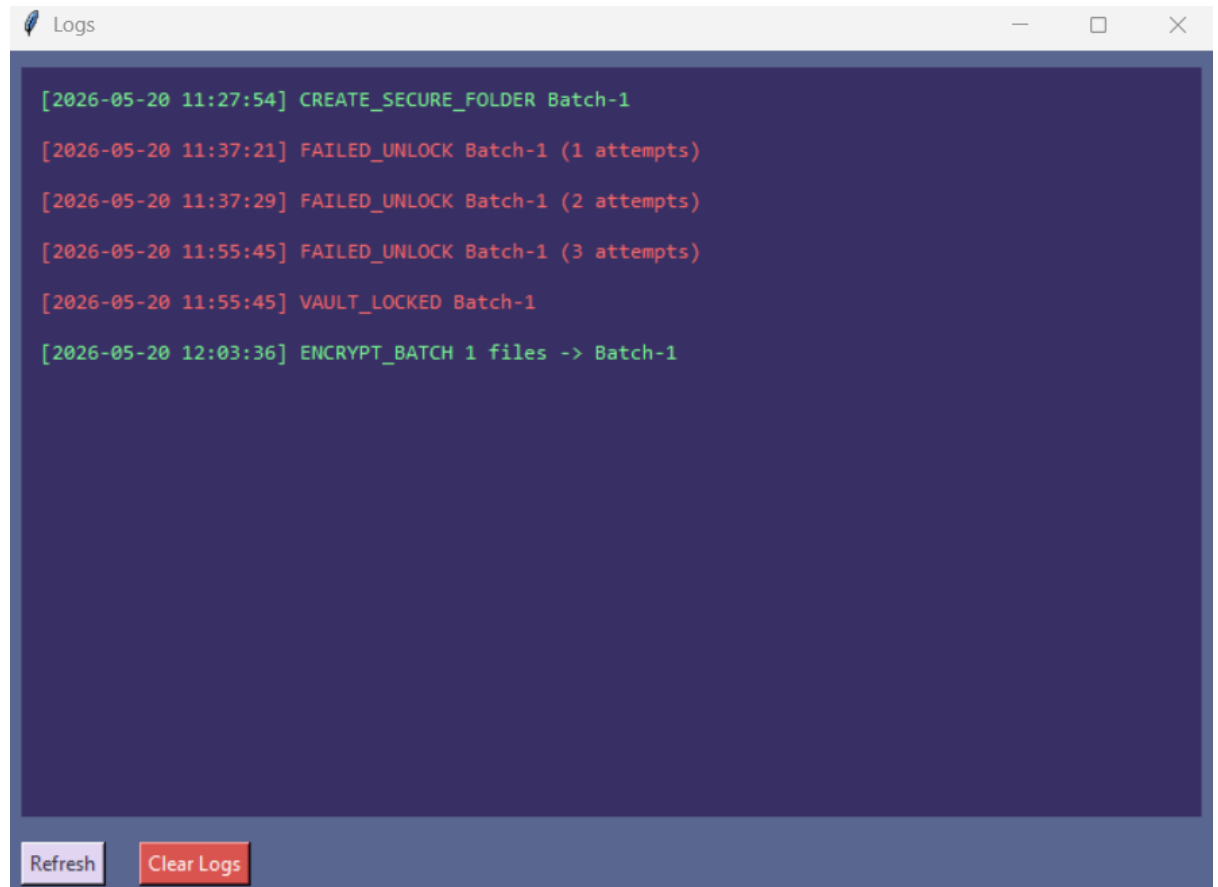
4. Log Based Error Tracking

- Security-related failures should be monitored for auditing and intrusion tracking. Failed access attempts and security events are permanently recorded in system logs for monitoring and analysis.
- **Validation Logic:**

```
log_action(
```

```
LOGS_DIR,  
"FAILED_UNLOCK",  
folder  
)
```

- **Output Preview:**



The screenshot shows a window titled "Logs" with a dark blue background and light blue text. The log entries are as follows:

- [2026-05-20 11:27:54] CREATE_SECURE_FOLDER Batch-1
- [2026-05-20 11:37:21] FAILED_UNLOCK Batch-1 (1 attempts)
- [2026-05-20 11:37:29] FAILED_UNLOCK Batch-1 (2 attempts)
- [2026-05-20 11:55:45] FAILED_UNLOCK Batch-1 (3 attempts)
- [2026-05-20 11:55:45] VAULT_LOCKED Batch-1
- [2026-05-20 12:03:36] ENCRYPT_BATCH 1 files -> Batch-1

At the bottom of the window, there are two buttons: "Refresh" and "Clear Logs".